Formalizing Fefferman's Official Periodic Navier–Stokes Problem (B) in Lean 4: Global Linear Strain Exclusion on $\mathbb{T}^3$, Biot–Savart Direction Depletion Identities, and Conditional Reductions via Beale–Kato–Majda

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Abstract

We present a self-contained Lean 4 formalization (`leanprover/lean4:v4.30.0-rc1`, Mathlib 4, 184 kernel-verified theorems across 5 modules with zero `sorry` and zero non-standard axioms outside `[propext, Classical.choice, Quot.sound]`) of Charles L. Fefferman's official periodic 3D incompressible Navier–Stokes problem statement (*Existence and Smoothness of the Navier–Stokes Equation*, Clay Mathematics Institute, 2006, Statement (B) on $\mathbb{R}^3/\mathbb{Z}^3$) together with unconditional geometric/kinematic identities and explicit **conditional reductions** to Beale–Kato–Majda continuation.

Scope and Honesty Statement: This work does **not** solve Clay Problem (B) or (A). Specifically:

1. We prove unconditionally that spatial $\mathbb{Z}^3$-periodicity excludes any non-zero *global* linear background strain $u_{\text{strain}}(x) = Ax$ on $\mathbb{R}^3$ (`z3_periodic_linear_eq_zero`), including the global uniaxial Burgers background strain $A = \text{diag}(-\alpha/2, -\alpha/2, \alpha)$ with $\alpha \neq 0$ (`burgersLinearStrain_not_z3_periodic`). Crucially, torus periodicity rules out only a *global* linear background strain: localized Burgers-like stretched vortex structures can still form locally on $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$, so global linear strain exclusion does **not** yield unconditional Clay (B).
2. We prove unconditionally the pointwise Biot–Savart direction-alignment depletion identity (`biot_savart_det_aligned_eq_zero`, `aligned_vorticity_stretching_integrand_zero`, `biot_savart_det_difference_identity`), the parabolic scale-invariance identities of 3D Navier–Stokes under $u_\lambda(x,t) = \lambda u(\lambda x, \lambda^2 t)$ (`navier_stokes_vorticity_dt_scaling_exponent`, `navier_stokes_scaling_homogeneity_deg6`, `unnormalized_quartic_scaling_mismatch`), and the cubic comparison ODE ceiling $M(t) \leq \max(M(0), C_{\text{CZ}}/\kappa'')$ (`comparison_ode_ceiling`) whenever a differentiable trajectory satisfies $M'(t) \leq C_{\text{CZ}} M(t)^2 - \kappa'' M(t)^3$.
3. We formulate the analytic closure hypotheses `PeriodicDirectionDepletionBound` ν, `PeriodicTetherClosingBound` ν, and `PeriodicBKMContinuation` ν as explicit `Prop`-valued hypotheses (never as axioms) and prove the **conditional reductions** `clay_B_exact_of_periodic_tet` and `clay_B_exact_of_torus_direction_depletion`. Whether arbitrary smooth periodic Navier–Stokes flows on $\mathbb{T}^3$ satisfy `PeriodicDirectionDepletionBound` ν or `PeriodicTetherClosingBound` ν remains an **open problem**.

All Lean 4 proofs are audited in-kernel by `AxiomAudit.lean` and attested with an Ed25519-signed `titan-audit` certificate and `sealguard` Merkle-root provenance manifest.

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1 Introduction and Explicit Proved vs. Open Status

In his official Clay Mathematics Institute problem description [1], Charles L. Fefferman formulates the periodic existence and smoothness problem (Statement (B)) for the three-dimensional incompressible Navier–Stokes equations on $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$. Formalizing PDE regularity statements in interactive theorem provers requires extreme care to avoid specification gaming, vacuous hypotheses, or conflating conditional reductions with unconditional proofs.

In this paper and its companion Lean 4 repository (`BenFrohman/fefferman-clay-navier-stokes`), we maintain a strict, machine-checked separation between:

- **Unconditionally proved theorems in Lean 4** (verified from Mathlib's standard axioms `[propext, Classical.choice, Quot.sound]` with zero `sorry` and zero custom `axiom` declarations), and
- **Open analytic hypotheses** (`PeriodicDirectionDepletionBound` ν, `PeriodicTetherClosingBound` ν and the classical PDE continuation package `PeriodicBKMContinuation` ν), which are defined as `Prop`s and passed explicitly as hypotheses to our conditional reduction theorems.

Table 1 provides a complete, transparent accounting of what is unconditionally proved and what remains open.

2 Fefferman's Official Periodic Navier–Stokes Formulation on $\mathbb{R}^3/\mathbb{Z}^3$

Let e_1, e_2, e_3 denote the standard orthonormal basis of $\mathbb{R}^3$. Following Fefferman [1], a velocity field $u : \mathbb{R}^3 \to \mathbb{R}^3$ or time-dependent field $u : \mathbb{R}^3 \times [0,\infty) \to \mathbb{R}^3$ and pressure $p : \mathbb{R}^3 \times [0,\infty) \to \mathbb{R}$ are spatial $\mathbb{Z}^3$-periodic (`IsZ3Periodic` / `IsZ3PeriodicScalar`) if for all $x \in \mathbb{R}^3$, $t \geq 0$, and $1 \leq j \leq 3$,

$$u(x + e_j, t) = u(x, t), \qquad p(x + e_j, t) = p(x, t). \tag{1}$$

The incompressible 3D Navier–Stokes system with viscosity $\nu > 0$ and zero external force ($f \equiv 0$) reads:

$$\frac{\partial u_i}{\partial t} + \sum_{j=1}^{3} u_j \frac{\partial u_i}{\partial x_j} = \nu \Delta u_i - \frac{\partial p}{\partial x_i} \qquad (x \in \mathbb{R}^3,\ t \geq 0,\ 1 \leq i \leq 3), \tag{2}$$

$$\operatorname{div} u = \sum_{i=1}^{3} \frac{\partial u_i}{\partial x_i} = 0 \qquad (x \in \mathbb{R}^3,\ t \geq 0), \tag{3}$$

$$u(x, 0) = u^{\circ}(x) \qquad (x \in \mathbb{R}^3). \tag{4}$$

Table 1: Summary of Unconditionally Proved Theorems vs. Open Analytic Hypotheses in `FeffermanClayNS`.

Declaration / Mathematical Item	**Status**	**Mathematical Meaning & Scope**
`Clay_B_exact` ν, `clay_B_exact_of_with_energy`	**PROVED** (Unconditional)	Exact encoding of Fefferman's Statement (B) (eqs. (1)–(3), (5), (7), (10)–(11) at $f \equiv 0$) and logical implication from the energy-bounded variant.
`library_huniq_unsatisfiable_at_zero`, `stepField_not_physical`, `repaired_hInt`, `repaired_huniq_at_zero`	**PROVED** (Unconditional)	Proof that unguarded local uniqueness fails against step-jump fields $u_{\text{step}}(t)$, and that the time-continuity repair in `IsPhysicalNSSolution` eliminates step-jump pathology and is satisfied by $u \equiv 0$.
`z3_periodic_linear_eq_zero`, `burgersLinearStrain_not_z3_periodic`	**PROVED** (Unconditional)	Any $\mathbb{Z}^3$-periodic linear map on $\mathbb{R}^3$ vanishes identically; the global linear Burgers strain $\mathrm{diag}(-\alpha/2, -\alpha/2, \alpha)x$ ($\alpha \neq 0$) is not $\mathbb{Z}^3$-periodic. (*Note:* Rules out global linear strain only; local Burgers-like vortex structures can still form on $\mathbb{T}^3$.)
`biot_savart_det_aligned_eq_zero`, `aligned_vorticity_stretching_integrand_zero`, `biot_savart_det_difference_identity`	**PROVED** (Unconditional)	Exact pointwise vanishing of the Biot–Savart stretching determinant $\det(\omega(y), z, \omega(x))$ when $\omega(y) = c\,\omega(x)$, reducing the stretching kernel to direction increments $\omega(y) - \omega(x)$.
`navier_stokes_vorticity_dt_scaling_exponent`, `navier_stokes_scaling_homogeneity_degree`, `unnormalized_quartic_scaling_mismatch`	**PROVED** (Unconditional)	Under parabolic scaling $u_\lambda(x,t) = \lambda u(\lambda x, \lambda^2 t)$, the scale-invariant cubic dissipation $\nu\lvert\omega\rvert^2\lvert\nabla\hat{\xi}\rvert^2 \sim \lambda^6$ matches $\partial_t\lvert\omega\rvert^2 \sim \lambda^6$, whereas unnormalized quartic terms scale as λ^8.
`unmodified_max_rate_of_cubic_source`, `viscous_direction_dissipation_bridge`, `scale_invariant_cubic_rate_of_direction_coercivity`	**PROVED** (Unconditional)	Algebraic and differential reduction from high-vorticity direction gradient coercivity $\lvert\nabla\hat{\xi}\rvert^2 \geq c_{\text{dir}}\lvert\omega\rvert$ to the cubic vorticity inequality $M' \leq C_{\text{CZ}}M^2 - \kappa'' M^3$ with $\kappa'' = \nu c_{\text{dir}}$.
`comparison_ode_ceiling`, `periodicTetherClosingBound_of_directionDepletion`	**PROVED** (Unconditional)	First-exit-time ODE comparison proof that $M'(t) \leq C_{\text{CZ}}M(t)^2 - \kappa'' M(t)^3$ on $[0,T]$ implies $M(t) \leq \max(M(0), C_{\text{CZ}}/\kappa'')$ for all $t \in [0,T]$.
`PeriodicDirectionDepletionBound` ν	**OPEN HYPOTHESIS**	Whether every smooth periodic Navier–Stokes flow on $\mathbb{T}^3$ satisfies $M'(t) \leq C_{\text{CZ}}M(t)^2 - \kappa'' M(t)^3$ along its existence horizon.
`PeriodicTetherClosingBound` ν	**OPEN HYPOTHESIS**	Whether every smooth periodic Navier–Stokes flow on $\mathbb{T}^3$ obeys a uniform L^∞ vorticity ceiling $\lVert\omega(t)\rVert_\infty \leq Y(u_0, \nu)$.
`PeriodicBKMContinuation` ν	**OPEN FORMALIZATION HYPOTHESIS**	Classical PDE continuation theorem (Kato 1972 + Beale–Kato–Majda 1984 on $\mathbb{T}^3$) stated as an explicit hypothesis in Lean 4.
`clay_B_exact_of_periodic_tether_bound`, `clay_B_exact_of_torus_direction_depletion`	**PROVED CONDITIONAL REDUCTIONS**	Conditional theorems proving that *if* the open hypotheses above hold, then `Clay_B_exact` ν follows.

In `FeffermanClayNS/FeffermanStatements.lean`, given smooth, divergence-free, $\mathbb{Z}^3$-periodic initial data $u^\circ \in C^\infty(\mathbb{R}^3/\mathbb{Z}^3)$ (`IsPeriodicSmoothInitialData`), a global smooth periodic solution (`IsPeriodicSmoothSolution`) satisfies (1)–(4) together with joint spacetime smoothness

$$p, u_i \in C^\infty(\mathbb{R}^3 \times [0,\infty)) \qquad (\texttt{ContDiff}\ \mathbb{R}\ \top\ (\texttt{uncurry}\ u)). \tag{5}$$

Fefferman's Statement (B) is encoded verbatim as `Clay_B_exact` ν, together with the energy-bounded variant `Clay_B_exact_with_energy` ν incorporating $\int_{\mathbb{T}^3} |u(x,t)|^2\,dx < C$ and the proved reduction `clay_B_exact_of_with_energy`.

2.1 Eliminating Step-Jump Vacuity in Local Existence Horizons

A subtle pitfall in formalizing PDE existence horizons in Lean 4 is that a pointwise time derivative predicate $\partial_t u(t,x) = \cdots$ without continuity in t permits discontinuous step-jump fields such as

$$u_{\text{step}}(t,x) = \begin{cases} (0,0,0) & \text{if } t \le 0, \\ (1,0,0) & \text{if } t > 0, \end{cases}$$

which satisfy $\partial_t u_{\text{step}}(t,x) = 0 = \nu\Delta u_{\text{step}} - \nabla p$ for every $t > 0$ while violating uniqueness from $u^\circ \equiv 0$. In `FeffermanStatements.lean`, we prove:

1. `library_huniq_unsatisfiable_at_zero`: An unguarded local uniqueness hypothesis on `existenceTimes_unpatched` is logically falsified ('False') at $u^\circ \equiv 0$ by u_{step}.
2. `stepField_not_physical`: Upgrading the solution predicate to `IsPhysicalNSSolution` (requiring time continuity `ContinuousOn (fun t => u t x) (Set.Icc 0 T)` and spatial boundedness `BddAbove`) rules out u_{step}.
3. `repaired_hInt`, `repaired_huniq_shift_at_zero`, and `repaired_huniq_at_zero`: The repaired physical solution class is non-vacuously inhabited by the trivial flow $u \equiv 0$ and satisfies local uniqueness via the energy Grönwall identity.

3 Global Linear Strain Exclusion on $\mathbb{T}^3$ and Its Exact Scope

On all of $\mathbb{R}^3$, the classic Burgers vortex lives inside an unbounded uniaxial linear background strain

$$u_{\text{strain}}(x) = Ax = \begin{pmatrix} -\alpha/2 & 0 & 0 \\ 0 & -\alpha/2 & 0 \\ 0 & 0 & \alpha \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \qquad \operatorname{tr}(A) = -\frac{\alpha}{2} - \frac{\alpha}{2} + \alpha = 0. \tag{6}$$

In `FeffermanClayNS/TorusStrainExclusion.lean`, we prove that any $\mathbb{Z}^3$-periodic linear map on $\mathbb{R}^3$ vanishes identically:

Theorem 3.1 (`z3_periodic_linear_eq_zero` and `burgersLinearStrain_not_z3_periodic`)**.** *Let $L : \mathbb{R}^3 \to_{\mathbb{R}} \mathbb{R}^3$ be any $\mathbb{R}$-linear map satisfying* IsZ3Periodic(L)*, i.e., $L(x + e_j) = L(x)$ for all $x \in \mathbb{R}^3$ and $j \in \{1,2,3\}$. Then $L = 0$. In particular, for every $\alpha \neq 0$, the divergence-free global Burgers linear background strain (6) satisfies* $\neg$IsZ3Periodic(*`burgersLinearStrain`* α)*.*

Proof. By linearity and periodicity at $x = 0$, $L(e_j) = L(0 + e_j) - L(0) = 0$ for each standard basis vector e_j. Since $\{e_1, e_2, e_3\}$ spans $\mathbb{R}^3$, $L(x) = \sum_{j=1}^3 x_j L(e_j) = 0$ for all $x \in \mathbb{R}^3$. For $A = \operatorname{diag}(-\alpha/2, -\alpha/2, \alpha)$, evaluation at e_3 yields $(Ae_3)_3 = \alpha \neq 0$. □

Remark 3.2 (Crucial Scope Warning: Why Global Linear Strain Exclusion Does Not Solve Clay (B))**.** Theorem `burgersLinearStrain_not_z3_periodic` rules out only a *global* linear background strain $u(x) = Ax$ across $\mathbb{R}^3/\mathbb{Z}^3$. On the compact torus $\mathbb{T}^3$, a smooth periodic velocity field u with zero mean ($\int_{\mathbb{T}^3} u\,dx = 0$) can still exhibit a large, locally linear strain matrix $\nabla u(x_0) = A$ inside a small neighborhood $B_r(x_0) \subset \mathbb{T}^3$, with the return flow closing smoothly elsewhere in the periodic box. Consequently, localized Burgers-like vortex tubes and sheets can still form locally on $\mathbb{T}^3$, and ruling out global linear strain does **not** yield an unconditional proof of Clay (B).

4 Biot–Savart Direction Alignment Depletion and Parabolic Scaling Identities

Taking the curl of (2) yields the 3D vorticity transport equation for $\omega = \nabla \times u$:

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega, \tag{7}$$

and the pointwise enstrophy evolution along $\omega = |\omega|\hat{\xi}$ ($|\hat{\xi}| = 1$ on $\{\omega \neq 0\}$):

$$\frac{1}{2}(\partial_t + u \cdot \nabla)|\omega|^2 = S(x)\omega(x) \cdot \omega(x) + \nu \Delta \frac{|\omega|^2}{2} - \nu |\nabla |\omega||^2 - \nu |\omega|^2 |\nabla \hat{\xi}|^2. \tag{8}$$

Under the Biot–Savart law, the vortex stretching quadratic form at x has integrand proportional to the 3×3 determinant

$$D(\omega(y), z, \omega(x)) = \det(\omega(y),\ z,\ \omega(x)) = (\omega(y) \times z) \cdot \omega(x), \qquad z = x - y.$$

In `TorusStrainExclusion.lean`, we prove three unconditional algebraic identities for this kernel:

1. **Collinear alignment vanishing (`biot_savart_det_aligned_eq_zero`):** Whenever $\omega(y) = c\,\omega(x)$ for any scalar $c \in \mathbb{R}$, $\det(c\,\omega(x), z, \omega(x)) = 0$.

2. **Complete stretching integrand vanishing under alignment (`aligned_vorticity_stretching_in`** The full singular stretching integrand $3\frac{(z\cdot\omega(x))\ \det(\omega(y),z,\omega(x))}{|z|^5}$ vanishes identically whenever $\omega(y) = c\,\omega(x)$.

3. **Exact direction-difference decomposition (`biot_savart_det_difference_identity`):** For all $\omega_x, \omega_y, z \in \mathbb{R}^3$,
$$\det(\omega_y, z, \omega_x) = \det(\omega_y - \omega_x, z, \omega_x), \tag{9}$$
demonstrating that vortex stretching at x is driven exclusively by deviations of $\omega(y)$ from the direction of $\omega(x)$ (Constantin–Fefferman geometric depletion [3]).

Furthermore, under parabolic Navier–Stokes rescaling $u_\lambda(x,t) = \lambda u(\lambda x, \lambda^2 t)$, the vorticity scales as $\omega_\lambda(x,t) = \lambda^2 \omega(\lambda x, \lambda^2 t)$ and the direction gradient scales as $(\nabla\hat{\xi})_\lambda(x,t) = \lambda(\nabla\hat{\xi})(\lambda x, \lambda^2 t)$. We prove in `navier_stokes_scaling_homogeneity_deg6` and `unnormalized_quartic_scaling_mismatch` that the viscous direction-dissipation density $\nu|\omega|^2|\nabla\hat{\xi}|^2$ scales homogeneously of degree 6 in λ:

$$\nu(\lambda^2|\omega|)^2(\lambda|\nabla\hat{\xi}|)^2 = \lambda^6 \nu |\omega|^2 |\nabla\hat{\xi}|^2, \tag{10}$$

matching $\partial_t|\omega_\lambda|^2 \sim \lambda^6$, whereas an unnormalized quartic damping $\kappa|\omega|^4$ scales as $\lambda^8 \neq \lambda^6$ for $\lambda \notin \{0, 1\}$.

5 Cubic Comparison ODE Ceiling and Conditional Reductions to `Clay_B_exact`

In `FeffermanClayNS/HighSetDepletionAndBKM.lean`, we first prove the algebraic and differential bridges showing that if at the spatial maximum $M(t) = \|\omega(t)\|_\infty = |\omega(t, x_*(t))|$ the high-vorticity direction gradient satisfies the scale-invariant coercivity lower bound $|\nabla\hat{\xi}(x_*)|^2 \geq c_{\mathrm{dir}} M(t)$ with $c_{\mathrm{dir}} > 0$, then the viscous direction-dissipation term $-\nu M(t)^2 |\nabla\hat{\xi}(x_*)|^2 \leq -\nu c_{\mathrm{dir}} M(t)^3$ produces the cubic differential inequality (`scale_invariant_cubic_rate_of_direction_coercivity`):

$$M'(t) \leq C_{\mathrm{CZ}} M(t)^2 - \kappa'' M(t)^3, \qquad \kappa'' = \nu c_{\mathrm{dir}} > 0. \tag{11}$$

We then prove unconditionally the global ceiling theorem for any differentiable function $M : \mathbb{R} \to \mathbb{R}$ obeying (11):

Theorem 5.1 (`comparison_ode_ceiling`)**.** *Let $M : \mathbb{R} \to \mathbb{R}$ be differentiable at every $t \in [0, T]$ and satisfy $M'(t) \leq C_{\mathrm{CZ}} M(t)^2 - \kappa'' M(t)^3$ on $[0, T]$ with $C_{\mathrm{CZ}} > 0$ and $\kappa'' > 0$. Then for every $t \in [0, T]$,*

$$M(t) \leq \max\left(M(0),\, \frac{C_{\mathrm{CZ}}}{\kappa''}\right). \tag{12}$$

Proof. Let $Y = \max(M(0), C_{\mathrm{CZ}}/\kappa'') > 0$. For any $\varepsilon > 0$, define the constant barrier $B(s) = Y + \varepsilon$. At any contact point $s \in [0, t]$ where $M(s) = Y + \varepsilon$, we have $M(s) > C_{\mathrm{CZ}}/\kappa'' > 0$, so $\kappa'' M(s) > C_{\mathrm{CZ}}$ and thus

$$M'(s) \leq M(s)^2 (C_{\mathrm{CZ}} - \kappa'' M(s)) < 0 = B'(s).$$

Applying Mathlib's first-exit-time comparison lemma `image_le_of_deriv_right_lt_deriv_boundary` yields $M(t) \leq Y + \varepsilon$ for every $\varepsilon > 0$, and `le_of_forall_pos_lt_add` concludes $M(t) \leq Y$. □

Finally, we state the two open analytic hypotheses and the conditional reductions:

Open Analytic Hypothesis 5.2 (`PeriodicDirectionDepletionBound` ν — OPEN)**.** For every smooth, divergence-free, $\mathbb{Z}^3$-periodic initial datum u°, there exist $C_{\mathrm{CZ}}, \kappa'' > 0$ such that along any local smooth periodic solution on $[0, T]$, $M(t) = \|\omega(t)\|_\infty$ is differentiable and obeys $M'(t) \leq C_{\mathrm{CZ}} M(t)^2 - \kappa'' M(t)^3$ on $[0, T]$.

Open Analytic Hypothesis 5.3 (`PeriodicTetherClosingBound` ν — OPEN)**.** For every smooth, divergence-free, $\mathbb{Z}^3$-periodic initial datum u°, there exists $Y \geq 0$ such that $\|\omega(t)\|_\infty \leq Y$ for all $t \in [0, T]$ along any local smooth periodic solution on $[0, T]$.

Theorem 5.4 (Conditional Reductions: `periodicTetherClosingBound_of_directionDepletion`, `clay_B_exact_of_periodic_tether_bound`, and `clay_B_exact_of_torus_direction_depletion`)**.**

1. *`PeriodicDirectionDepletionBound` ν implies `PeriodicTetherClosingBound` ν with explicit ceiling $Y = \max(\|\omega_0\|_\infty, C_{\mathrm{CZ}}/\kappa'')$.*
2. *Conditional on `PeriodicTetherClosingBound` ν (OPEN) and `PeriodicBKMContinuation` ν (classical Kato + Beale–Kato–Majda continuation [4, 2]), `Clay_B_exact` ν holds.*
3. *Conditional on `PeriodicDirectionDepletionBound` ν (OPEN) and `PeriodicBKMContinuation` ν, `Clay_B_exact` ν holds.*

6 Verbatim Lean 4 Kernel Audit and Cryptographic Attestation

Compiling the repository with `lake build && lake env lean AxiomAudit.lean` executes an in-kernel traversal of the Lean 4 environment via `Lean.collectAxioms` across all 184 theorems in the 4 theorem-bearing modules and prints the exact axiom footprint of the key theorems:

Listing 1: Verbatim output of `lake env lean AxiomAudit.lean` on `fefferman-clay-navier-stokes`.

```
FeffermanClayNS.TorusStrainExclusion: theorems=19 nonstandard=0
FeffermanClayNS.HighSetDepletionAndBKM: theorems=9 nonstandard=0
FeffermanClayNS.FeffermanStatements: theorems=13 nonstandard=0
FeffermanClayNS.Defs: theorems=143 nonstandard=0
TOTAL theorems=184 nonstandard=0
'FeffermanClayNS.clay_B_exact_of_with_energy' depends on axioms: [propext, Classical.choice,
    Quot.sound]
'FeffermanClayNS.library_huniq_unsatisfiable_at_zero' depends on axioms: [propext, Classical.
    choice, Quot.sound]
'FeffermanClayNS.repaired_huniq_at_zero' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'FeffermanClayNS.z3_periodic_linear_eq_zero' depends on axioms: [propext, Classical.choice,
    Quot.sound]
'FeffermanClayNS.burgersLinearStrain_not_z3_periodic' depends on axioms: [propext, Classical.
    choice, Quot.sound]
'FeffermanClayNS.cross3_self' depends on axioms: [propext, Classical.choice, Quot.sound]
'FeffermanClayNS.det3_self_right' depends on axioms: [propext, Classical.choice, Quot.sound]
'FeffermanClayNS.biot_savart_det_aligned_eq_zero' depends on axioms: [propext, Classical.choice
    , Quot.sound]
'FeffermanClayNS.aligned_vorticity_stretching_integrand_zero' depends on axioms: [propext,
    Classical.choice, Quot.sound]
'FeffermanClayNS.biot_savart_det_difference_identity' depends on axioms: [propext, Classical.
    choice, Quot.sound]
'FeffermanClayNS.navier_stokes_vorticity_dt_scaling_exponent' depends on axioms: [propext,
    Classical.choice, Quot.sound]
'FeffermanClayNS.navier_stokes_scaling_homogeneity_deg6' depends on axioms: [propext, Classical
    .choice, Quot.sound]
'FeffermanClayNS.unnormalized_quartic_scaling_mismatch' depends on axioms: [propext, Classical.
    choice, Quot.sound]
'FeffermanClayNS.unmodified_max_rate_of_cubic_source' depends on axioms: [propext, Classical.
    choice, Quot.sound]
'FeffermanClayNS.viscous_direction_dissipation_bridge' depends on axioms: [propext, Classical.
    choice, Quot.sound]
'FeffermanClayNS.scale_invariant_cubic_rate_of_direction_coercivity' depends on axioms: [
    propext, Classical.choice, Quot.sound]
'FeffermanClayNS.comparison_ode_ceiling' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'FeffermanClayNS.periodicTetherClosingBound_of_directionDepletion' depends on axioms: [propext,
     Classical.choice, Quot.sound]
'FeffermanClayNS.clay_B_exact_of_periodic_tether_bound' depends on axioms: [propext, Classical.
    choice, Quot.sound]
'FeffermanClayNS.clay_B_exact_of_torus_direction_depletion' depends on axioms: [propext,
    Classical.choice, Quot.sound]
```

In addition to `AxiomAudit.lean`, the repository includes an Ed25519-signed `titan-audit` certificate (`audit/TITAN_AUDIT_CERTIFICATE.json`, DOI `10.5281/zenodo.23630648`) and a `sealguard` + `sentinel-gate` Merkle-root provenance and classified IP containment manifest (`audit/SEALGUARD_CLASSIFIED_MANIFEST.json`, DOIs `10.5281/zenodo.23631670` and `10.5281/zenodo.23632766`).

References

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